

A two-stage randomized dose optimization design for biomarker-targeted phase I oncology trials

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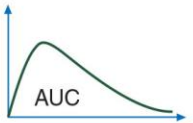
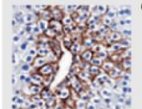
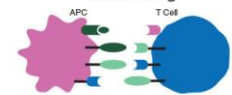
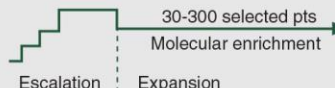
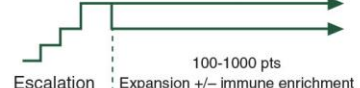
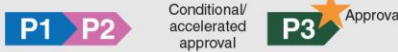
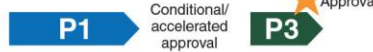


Alex Kaizer
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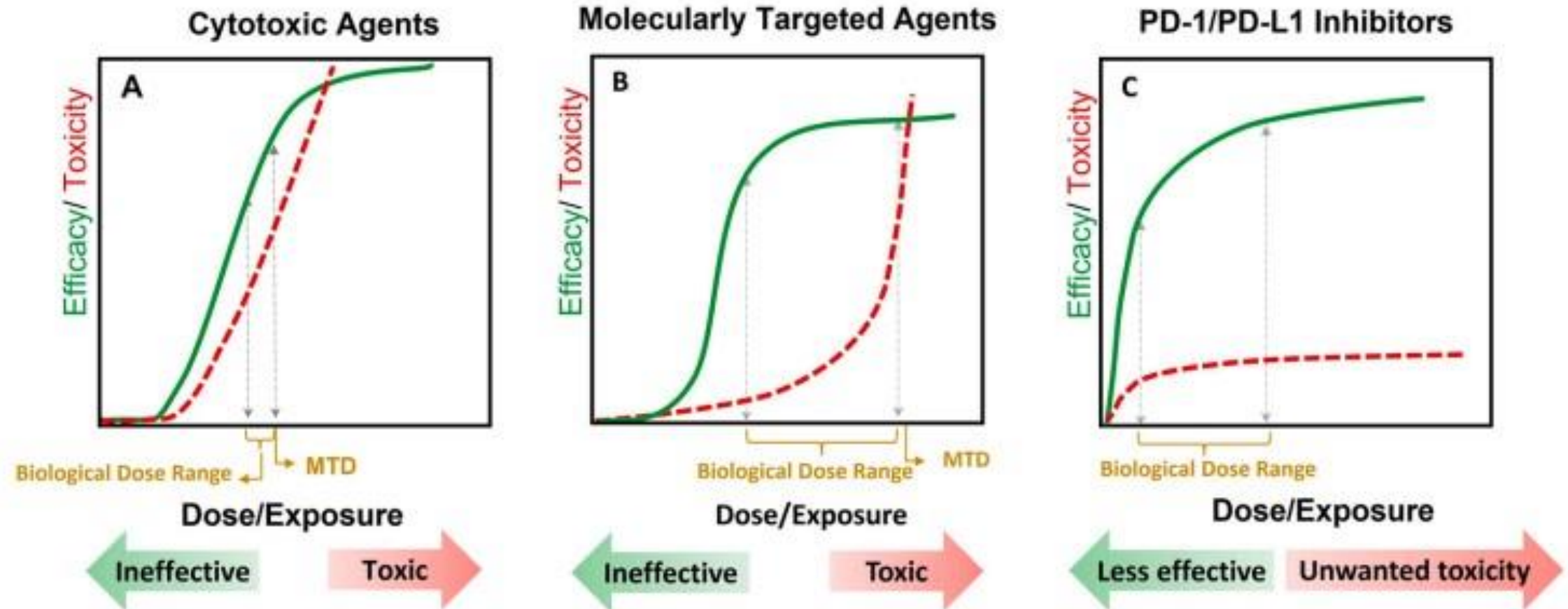


Brian Hobbs
Telperian

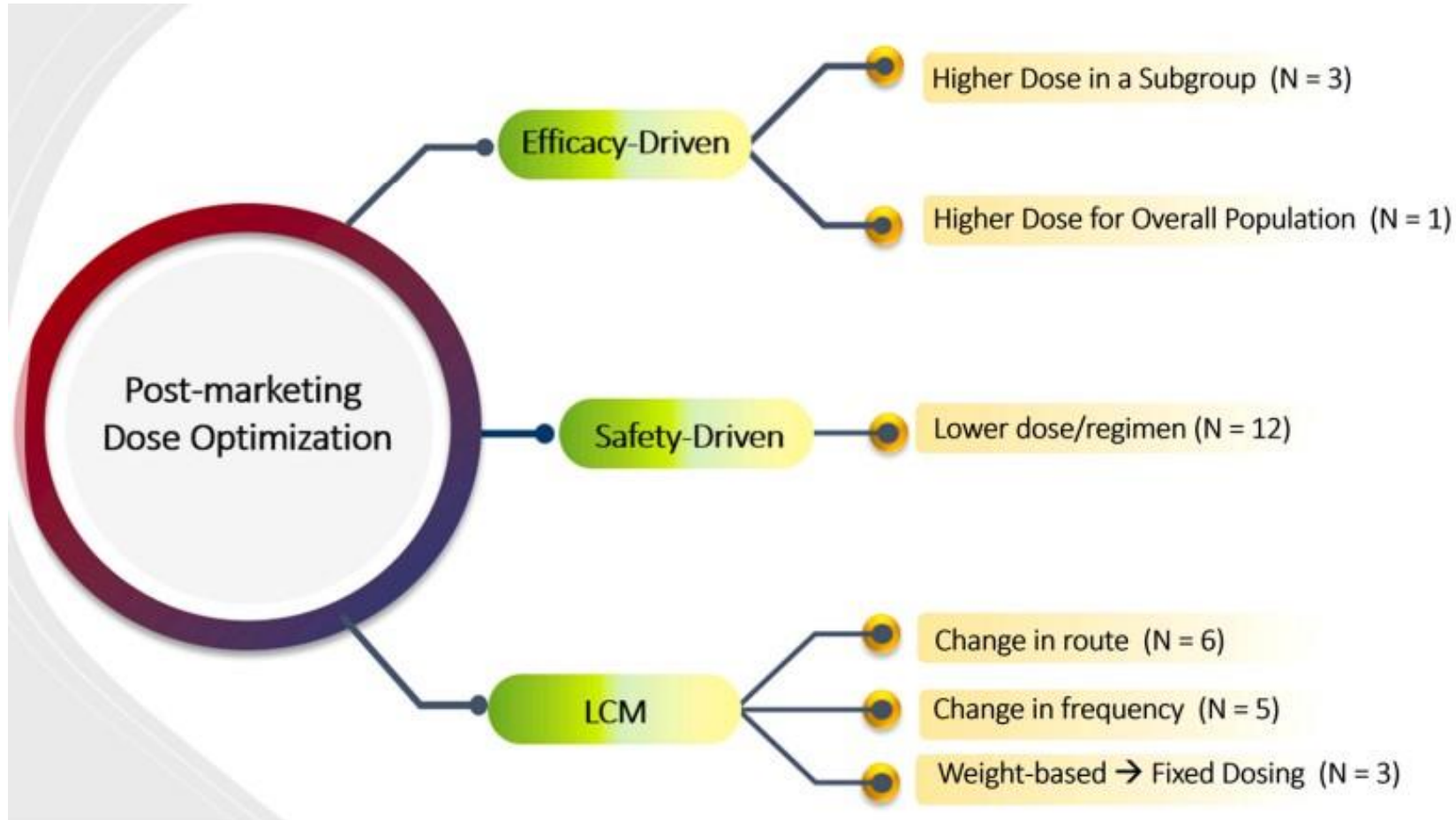
Accelerated approval has sped up the drug development process

	Cytotoxic chemotherapy	Molecularly targeted agents	Immuno-stimulatory antibodies
Patients number	 30-50 unselected patients	 30-200 molecularly selected patients	 100-1000 immunologically selected patients
Route of administration	IV > Oral 	Oral > IV 	Novel routes of administration (intra-tumoral) 
Toxicity	MTD quasi-systematically reached	MTD unconsciously reached	MTD rarely reached → MAD
PK/PD - biomarkers	Traditional PK limited PD 	Traditional PK with potential for PK - based dose recommendation Biomarker-driven PD for target assay validation and molecular enrichment 	PK and pD-based dose recommendation? repeated PD for dynamic biomarkers and immunological monitoring 
Design	Traditional 3 + 3 dose-escalation design 	3 + 3 dose-escalation design with large expansion cohorts in selected populations 	Accelerated titration/adaptive design multiple parallel expansion cohorts long-term follow-up + drug rechallenge 
Drug approval	Based on later phase 2 or 3 trials 	Conditional of accelerated approval based on large molecularly selected expansion cohorts 	Conditional of accelerated approval based on histology and immune-biomarker selected expansion cohorts 
Drug development timeframe	10 years	5-8 years	<5 years

MTD-focused dose selection may select a too-high dose for non-cytotoxic treatments



Post-marketing studies have often led to lower recommended doses



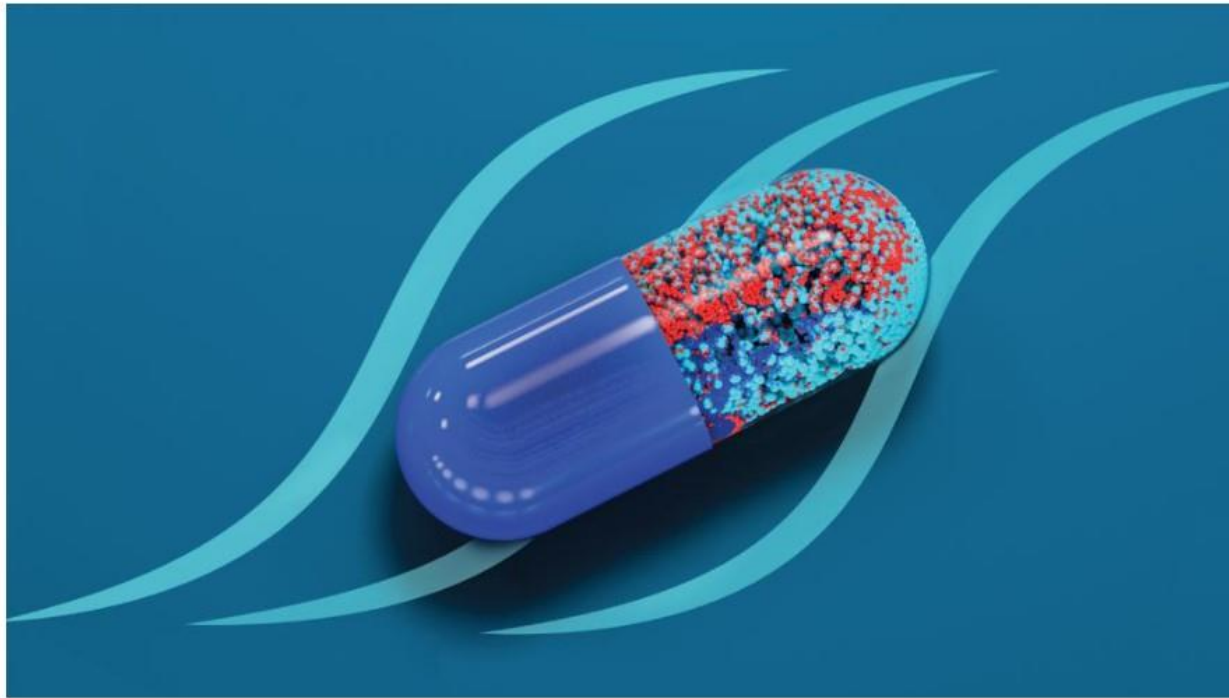
Recent examples show dose modifications to reduce toxicities or side effects

Drug	Initial Dose and Trials	Modified or Added Dose and Trials	Reason for Modified or Added Dose
Small-molecule drugs			
Ceritinib (Zykadia)	750 mg PO daily fasted (ASCEND-1)	450 mg PO daily with food (ASCEND-8)	Reduce gastrointestinal toxic effects
Dasatinib (Sprycel)	70 mg PO twice daily (CA180013, CA180005, CA180006, and CA180015)	100 mg PO daily (CA180034)	Reduce hematologic toxic effects and fluid retention
Niraparib (Zejula)	300 mg PO daily (NOVA)	200 mg PO daily (PRIMA)	Reduce thrombocytopenia in patients with a lower platelet count or lower body weight
Ponatinib (Iclusig)	45 mg PO daily (PACE)	45 mg PO daily, then 15 mg PO daily once $\leq 1\%$ BCR-ABL is achieved (OPTIC)	Reduce vascular occlusive events
Chemotherapy			
Cabazitaxel (Jevtana)	25 mg/m ² IV every 3 wk (TROPIC)	20 mg/m ² IV every 3 wk (PROSELICA)	Reduce hematologic toxic effects and infections
Antibody–drug conjugates			
Gemtuzumab ozogamicin (Mylotarg)	9 mg/m ² IV on days 1 and 15 (Study 201, Study 202, and Study 203)	3 mg/m ² IV on days 1, 4, and 7 (Mylofrance-1)	Reduce veno-occlusive disease and treatment-related mortality

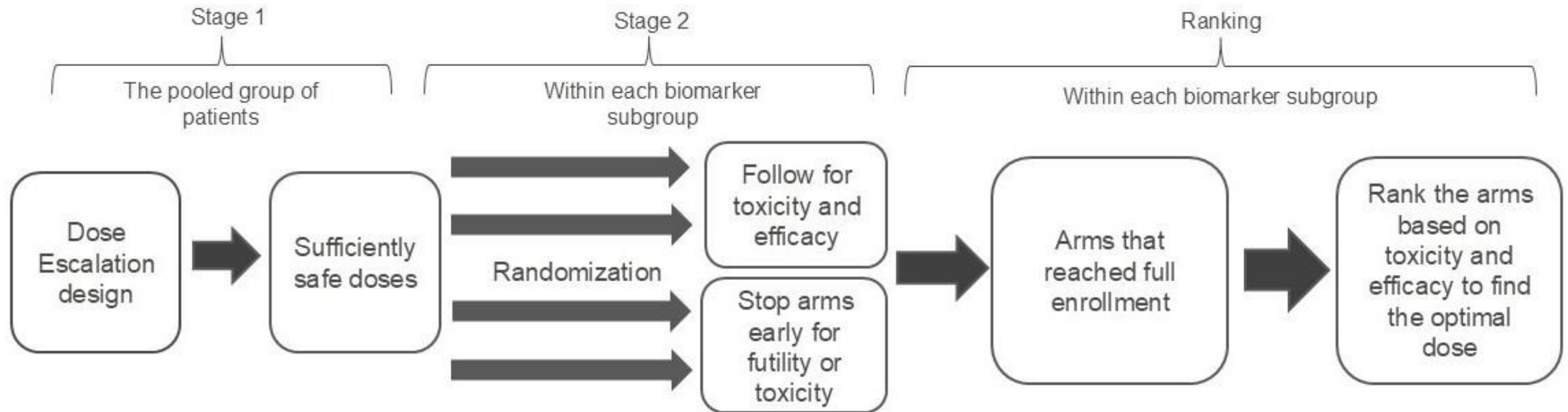
FDA's Project Optimus encourages randomization to a range of dose

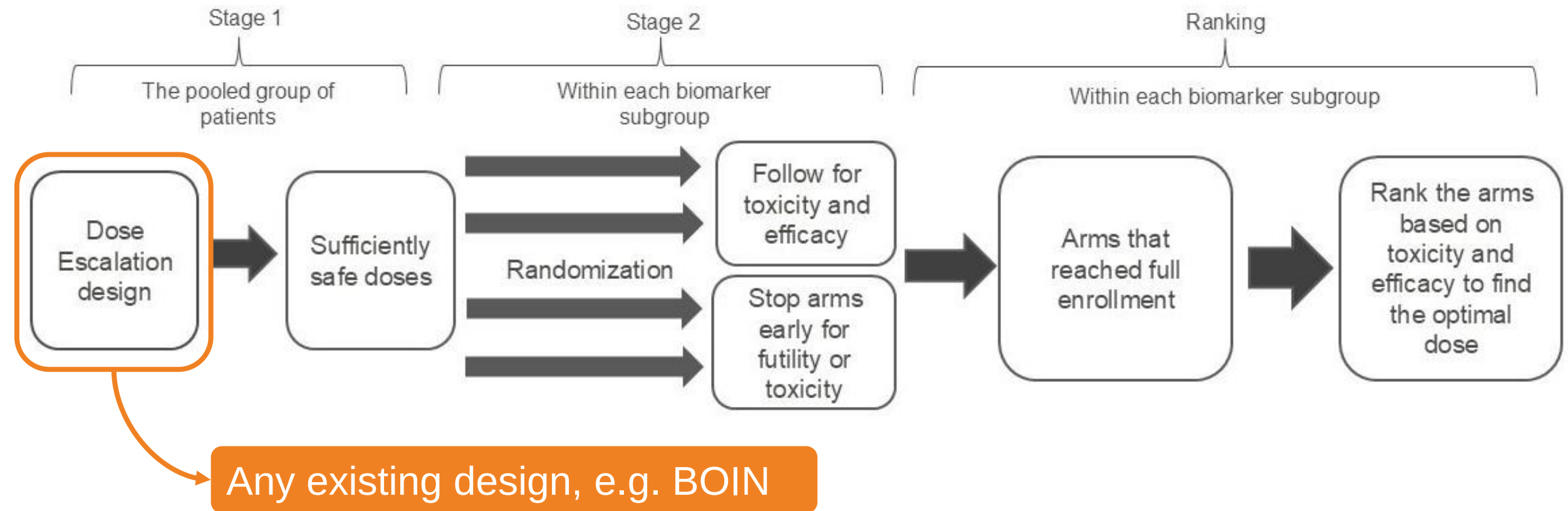
Project Optimus

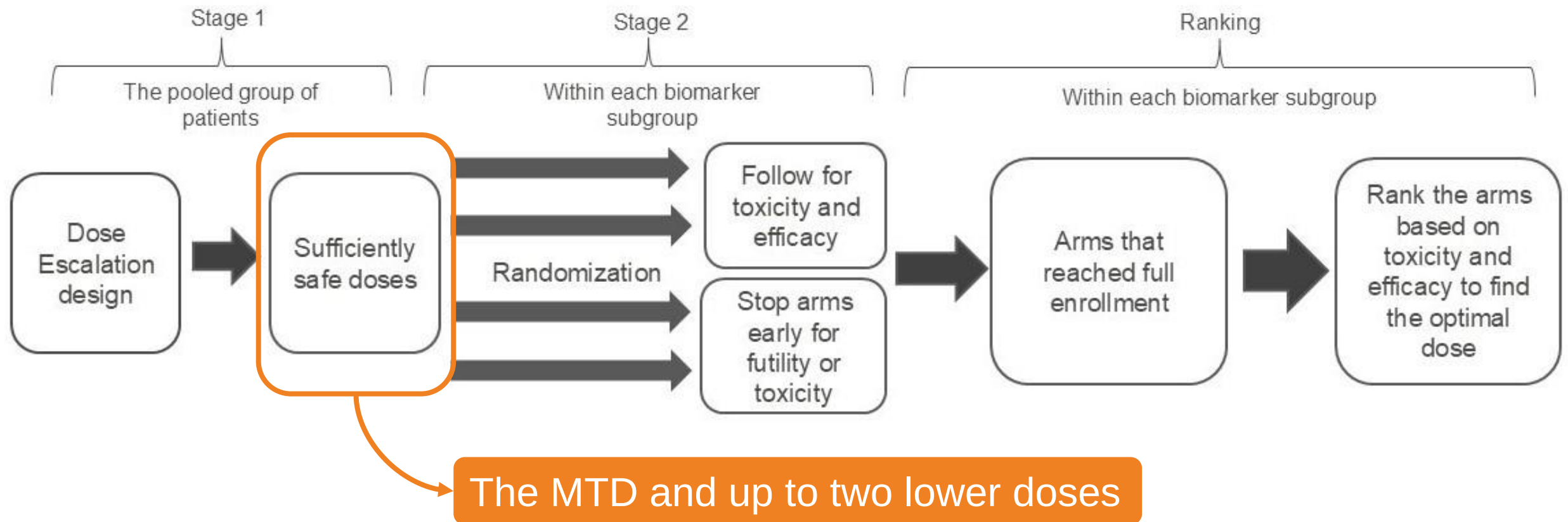
Reforming the dose optimization and dose selection paradigm in oncology

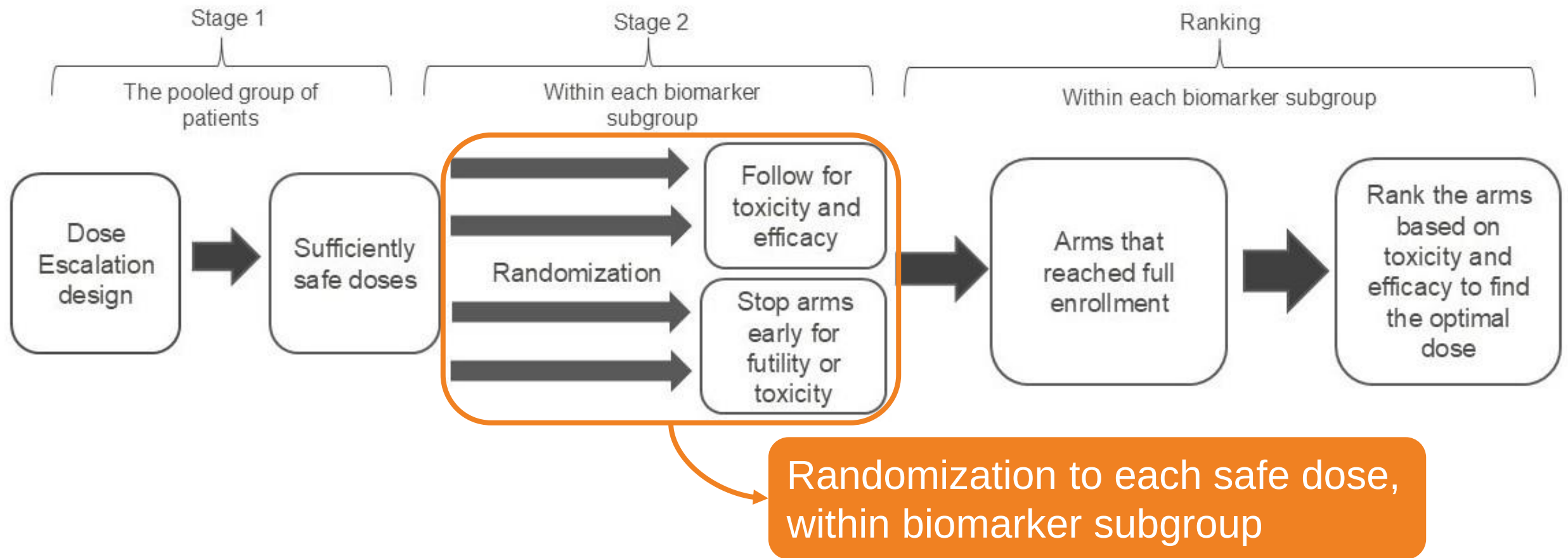


Proposed study design randomizes to safe doses within biomarker subgroups

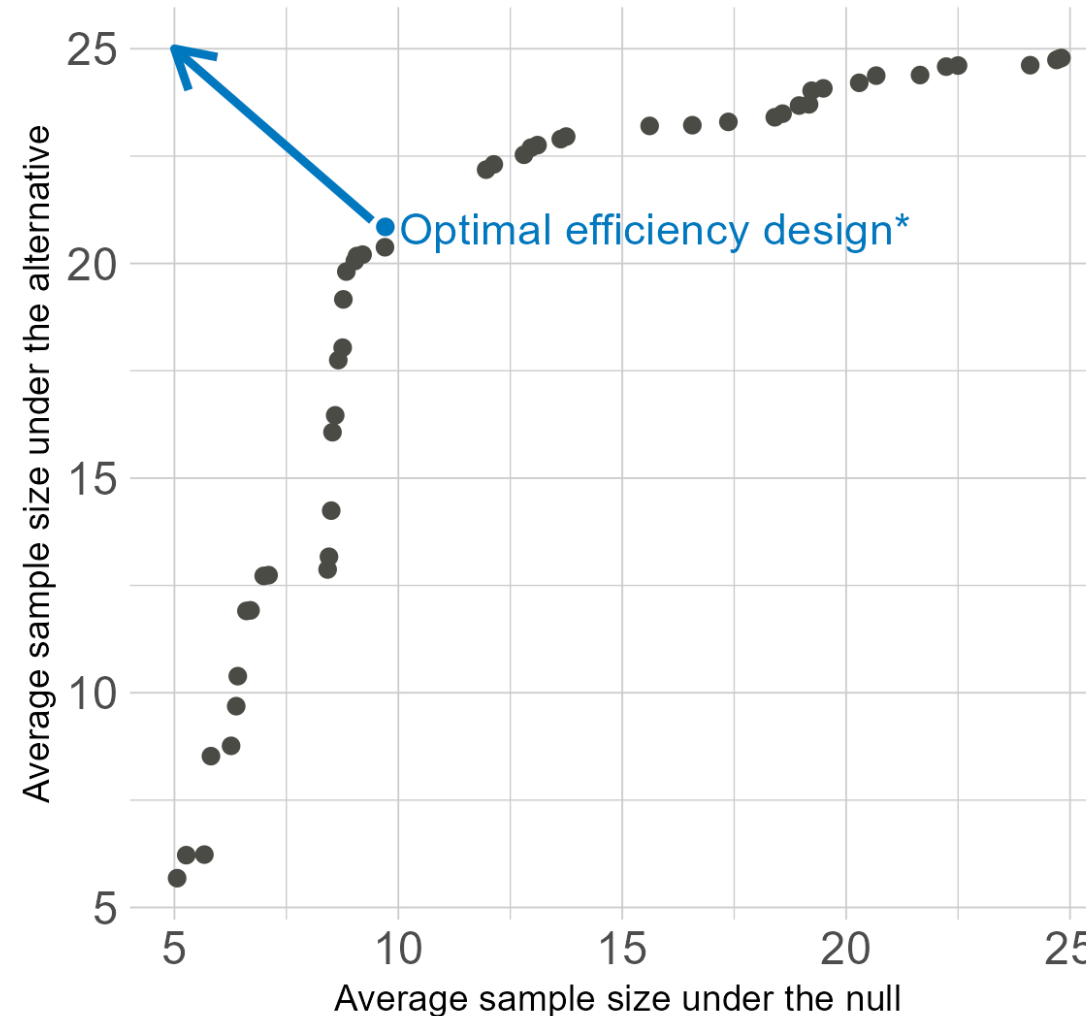


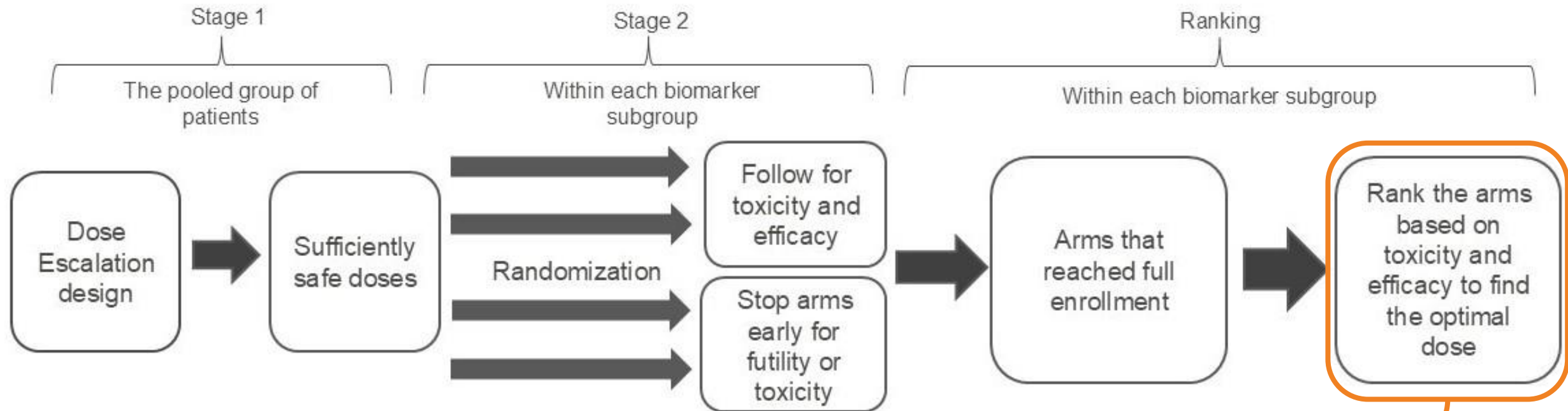






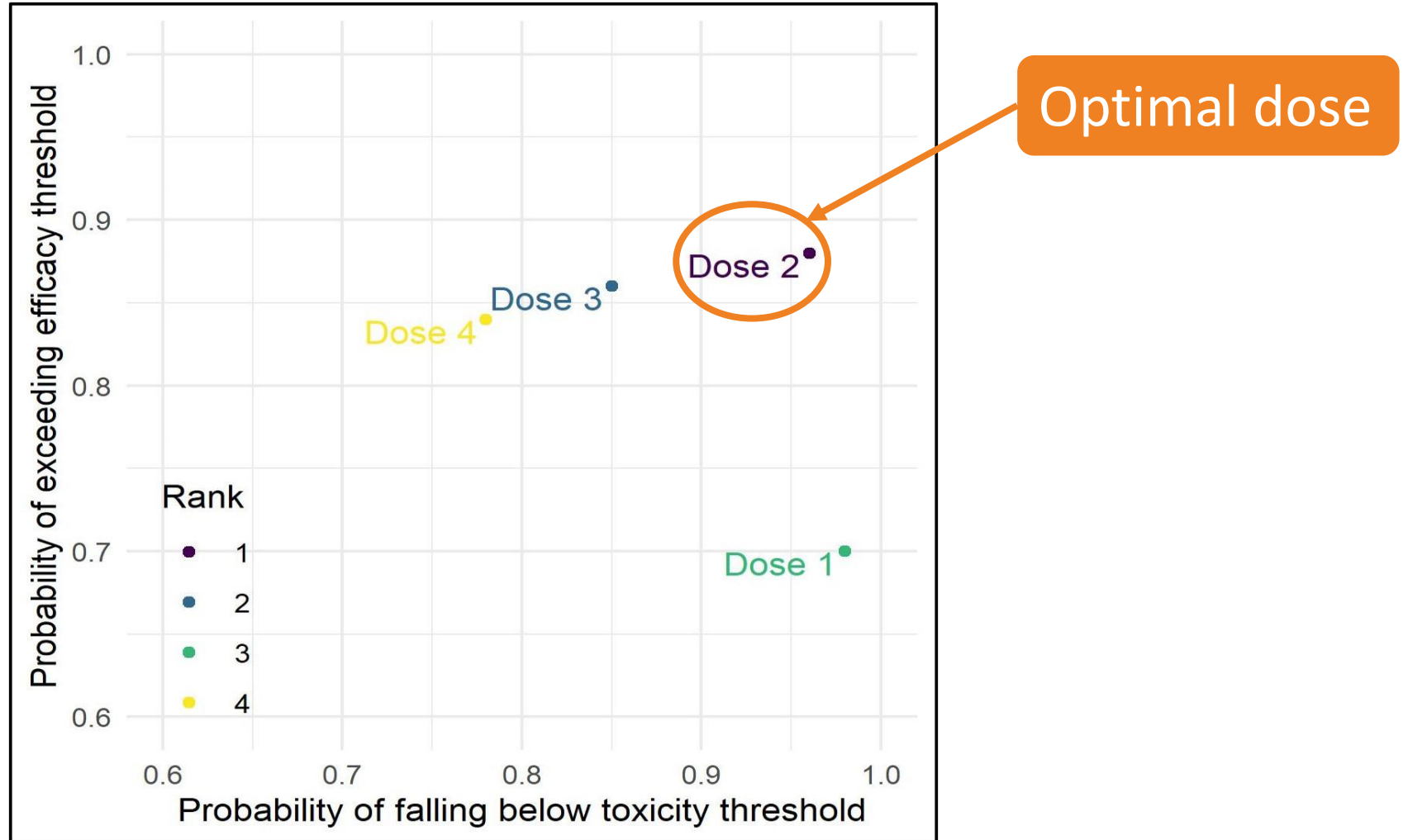
Interim monitoring with the optimal efficiency design





Ranking based on posterior probabilities

The optimal dose minimizes the Euclidean distance to the (1, 1) point



Ten scenarios averaged results over 1000 simulated trial datasets

- Cohorts of size 3 in stage 1, maximum sample size of 30
- Stage 1 considered both 3+3 and BOIN designs
- Sample size fixed at 20 per dosing arm in stage 2, with interim monitoring after every 5 patients
- Considered stage 2 designs with type I error in (0.05, 0.15) and power of at least 0.75
- Safety thresholds fixed at $\theta^S = 0.8$ and $\theta^{*S} = 0.2$
- Optimal efficiency efficacy thresholds of:
 - $\theta^e = 0.94$ and $\theta^{*e} = 0.2$ when $p_{0j} = 0.1$ and $p_{1j} = 0.3$
 - $\theta^e = 0.9$ and $\theta^{*e} = 0.15$ when $p_{0j} = 0.15$ and $p_{1j} = 0.4$

Scenario	Biomarker	True Efficacy rates, per dose level					True toxicity rates, per dose level					Null response rate used in the design	Alternative response rate used in the design
		1	2	3	4	5	1	2	3	4	5		
1	1	0.10	0.20	0.30	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.10	0.20	0.30	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
2	1	0.10	0.20	0.30	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.10	0.20	0.30	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.15	0.40
3	1	0.10	0.20	0.30	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.15	0.25	0.40	0.40	0.40	0.05	0.10	0.15	0.25	0.60	0.10	0.30
4	1	0.10	0.20	0.30	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.15	0.25	0.40	0.40	0.40	0.05	0.10	0.15	0.25	0.60	0.15	0.40
5	1	0.10	0.20	0.25	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.10	0.20	0.25	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
6	1	0.10	0.20	0.25	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.10	0.20	0.25	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.15	0.40
7	1	0.10	0.20	0.25	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.15	0.25	0.35	0.40	0.40	0.05	0.10	0.15	0.25	0.60	0.10	0.30
8	1	0.10	0.20	0.25	0.30	0.30	0.05	0.10	0.15	0.25	0.60	0.10	0.30
	2	0.15	0.25	0.35	0.40	0.40	0.05	0.10	0.15	0.25	0.60	0.15	0.40
9	1	0.10	0.20	0.20	0.40	0.40	0.05	0.10	0.10	0.25	0.60	0.10	0.30
	2	0.10	0.20	0.20	0.40	0.40	0.05	0.10	0.10	0.25	0.60	0.10	0.30
10	1	0.10	0.20	0.20	0.40	0.40	0.10	0.10	0.10	0.10	0.60	0.10	0.30
	2	0.10	0.20	0.20	0.40	0.40	0.10	0.10	0.10	0.10	0.60	0.10	0.30

Scenario	Proportion of times the optimal dose is selected as safe	
	3+3	BOIN
1	0.70	0.97
2	0.72	0.97
3	0.72	0.97
4	0.71	0.97
5	0.42	0.81
6	0.42	0.80
7	0.44	0.79
8	0.39	0.79
9	0.48	0.87
10	0.69	0.92

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6	0.42	0.80
7	0.44	0.79
8	0.39	0.79
9	0.48	0.87
10	0.69	0.92

BOIN identifies
the optimal
dose as safe at
much higher
rates than 3+3

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7	0.44	0.79
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Best performance
when optimal dose
is lower than MTD

Scenario	Proportion of times the optimal dose is selected as safe	
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7	0.44	0.79
8	0.39	0.79
9	0.48	0.87
10	0.69	0.92

3+3 often identifies
an MTD lower than
the true MTD,
missing the optimal
dose when they are
the same

Scenario	Proportion of times truly futile arms were terminated early for futility	Proportion of times truly toxic arms were terminated early for toxicity
1	0.45	0.84
2	0.38	0.81
3	0.38	0.87
4	0.44	0.90
5	0.34	0.82
6	0.37	0.88
7	0.34	0.86
8	0.33	0.86
9	0.39	0.93
10	0.38	0.92

*Stage 2 results based on BOIN

Scenario	Proportion of times truly futile arms were terminated early for futility	Proportion of times truly toxic arms were terminated early for toxicity
1	0.45	0.84
2	0.38	0.81
3	0.38	0.87
4	0.44	0.90
5	0.34	0.82
6	0.37	0.88
7	0.34	0.86
8	0.33	0.86
9	0.39	0.93
10	0.38	0.92

Truly toxic arms
were terminated
early with high
probability

*Stage 2 results based on BOIN

Scenario	Proportion of times the optimal dose is identified correctly	Proportion of times the identified optimal dose is lower than the true optimal dose	Proportion of times the identified optimal dose is higher than the true optimal dose	Proportion of times the identified optimal dose is lower than the identified MTD
1	0.53	0.33	0.14	0.79
2	0.52	0.33	0.16	0.76
3	0.49	0.38	0.13	0.83
4	0.53	0.33	0.14	0.78
5	0.15	0.85	0.00	0.79
6	0.17	0.83	0.00	0.76
7	0.15	0.85	0.00	0.77
8	0.18	0.82	0.00	0.76
9	0.22	0.78	0.00	0.76
10	0.63	0.37	0.00	0.50

Scenario	Proportion of times the optimal dose is identified correctly	Proportion of times the identified optimal dose is lower than the true optimal dose	Proportion of times the identified optimal dose is higher than the true optimal dose	Proportion of times the identified optimal dose is lower than the identified MTD
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7	0.15	0.85	0.00	0.77
8	0.18	0.82	0.00	0.76
9	0.22	0.78	0.00	0.76
10	0.63	0.37	0.00	0.50

When the true optimal dose is the same as the MTD, the identified dose is always equal to or lower than the true optimal dose

Scenario	Proportion of times the optimal dose is identified correctly	Proportion of times the identified optimal dose is lower than the true optimal dose	Proportion of times the identified optimal dose is higher than the true optimal dose	Proportion of times the identified optimal dose is lower than the identified MTD
1	0.53	0.33	0.14	0.79
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4	0.53	0.33	0.14	0.78
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6	0.17	0.83	0.00	0.76
7	0.15	0.85	0.00	0.77
8	0.18	0.82	0.00	0.76
9	0.22	0.78	0.00	0.76
10	0.63	0.37	0.00	0.50

When the true optimal dose is less toxic than the MTD, the identified dose is equal to or lower than the true optimal dose >80% of the time

Scenario	Proportion of times the optimal dose is identified correctly	Proportion of times the identified optimal dose is lower than the true optimal dose	Proportion of times the identified optimal dose is higher than the true optimal dose	Proportion of times the identified optimal dose is lower than the identified MTD
1	0.53	0.33	0.14	0.79
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8	0.18	0.82	0.00	0.76
9	0.22	0.78	0.00	0.76
10	0.63	0.37	0.00	0.50

The identified optimal dose is lower than the identified MTD >75% of the time in most scenarios

Scenario	Proportion of times the optimal dose is identified correctly	Proportion of times the identified optimal dose is lower than the true optimal dose	Proportion of times the identified optimal dose is higher than the true optimal dose	Proportion of times the identified optimal dose is lower than the identified MTD
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8	0.18	0.82	0.00	0.76
9	0.22	0.78	0.00	0.76
10	0.63	0.37	0.00	0.50

True MTD in scenario 10 had 10% toxicity; in close to 50% of simulated trials the identified optimal dose was the same as the identified MTD

Ceritinib case study exemplifies the importance of careful study design

Drug	Initial Dose and Trials	Modified or Added Dose and Trials	Reason for Modified or Added Dose
Small-molecule drugs			
Ceritinib (Zykadia)	750 mg PO daily fasted (ASCEND-1)	450 mg PO daily with food (ASCEND-8)	Reduce gastrointestinal toxic effects
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Chemotherapy			
Cabazitaxel (Jevtana)	25 mg/m ² IV every 3 wk (TROPIC)	20 mg/m ² IV every 3 wk (PROSELICA)	Reduce hematologic toxic effects and infections
Antibody–drug conjugates			
Gemtuzumab ozogamicin (Mylotarg)	9 mg/m ² IV on days 1 and 15 (Study 201, Study 202, and Study 203)	3 mg/m ² IV on days 1, 4, and 7 (Mylofrance-1)	Reduce veno-occlusive disease and treatment-related mortality

Poor properties when optimal dose much lower than the MTD

- 12 dosing arms
- Only 34% of simulated trials continued the true optimal dose to stage 2
- 4% of simulated trials identified the true optimal dose
- 82% of simulated trials identified a higher optimal dose than the truth
- 86% of simulated trials identified an optimal dose that was lower than the true MTD

Dose (mg/day)	True efficacy in crizotinib-pretreated group	True efficacy in the crizotinib-naive group	Toxicity rate
50	0.25	0.25	0.012
100	0.25	0.25	0.021
200	0.25	0.25	0.039
300	0.25	0.25	0.060
400	0.56	0.62	0.084
500	0.56	0.62	0.109
600	0.56	0.62	0.137
650	0.56	0.62	0.152
700	0.56	0.62	0.167
750	0.56	0.62	0.182
900	0.56	0.62	0.227
1050	0.56	0.62	0.270

True optimal dose

True MTD

In Summary

- Randomization within biomarker-specific subgroups addresses the goals of Project Optimus by exploring multiple doses
- Proposed design selects less toxic dose than MTD most of the time, with more information for later phases
- Bayesian sequential monitoring allows focus on promising doses by terminating futile and toxic doses
- Design matters – including dose selection and phase I escalation design

Thank you! Questions?

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